



THE FARMER'S FORECAST

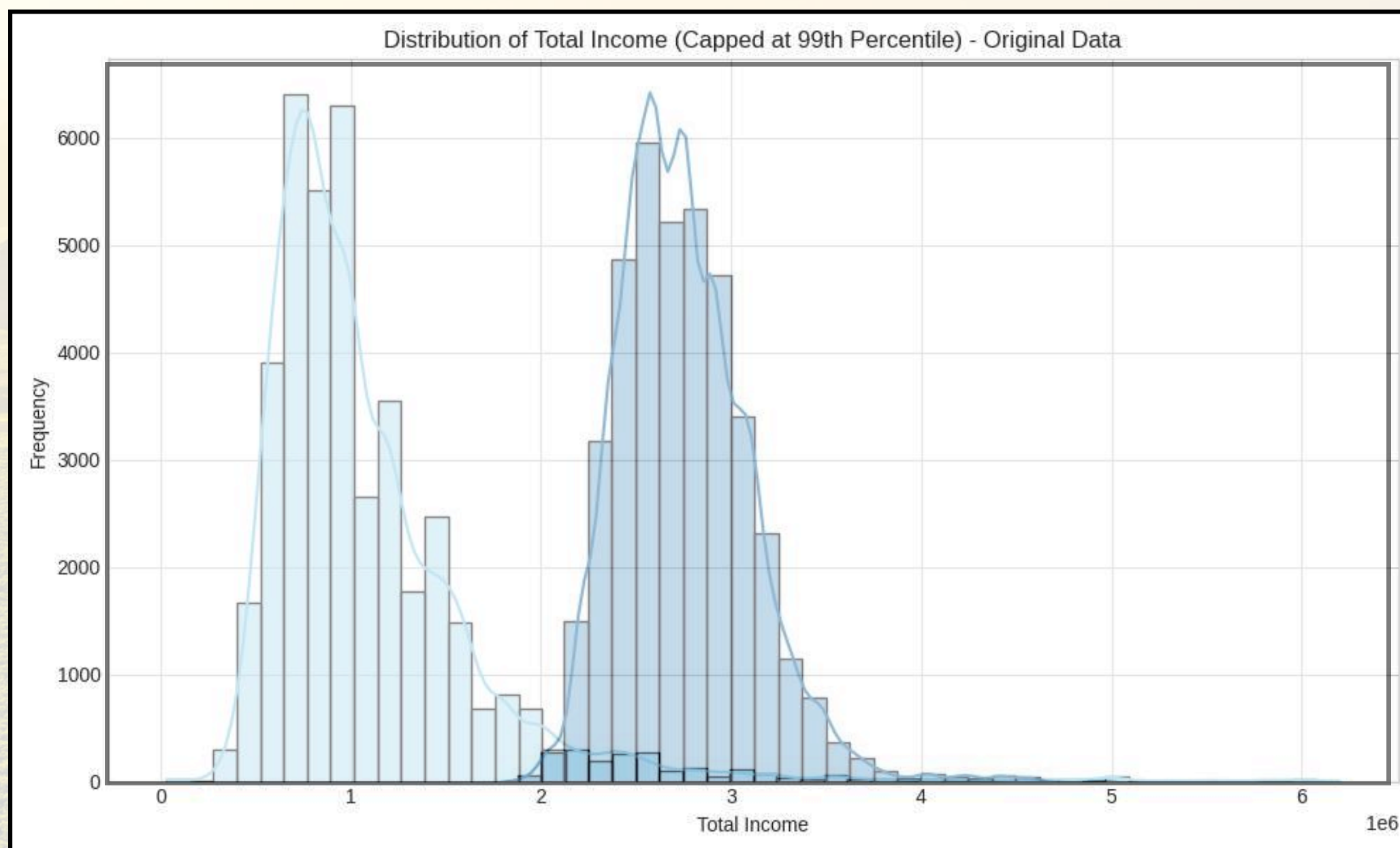
PREDICTING ANNUAL INCOME

A Predictive Modelling Approach for the Agricultural Sector

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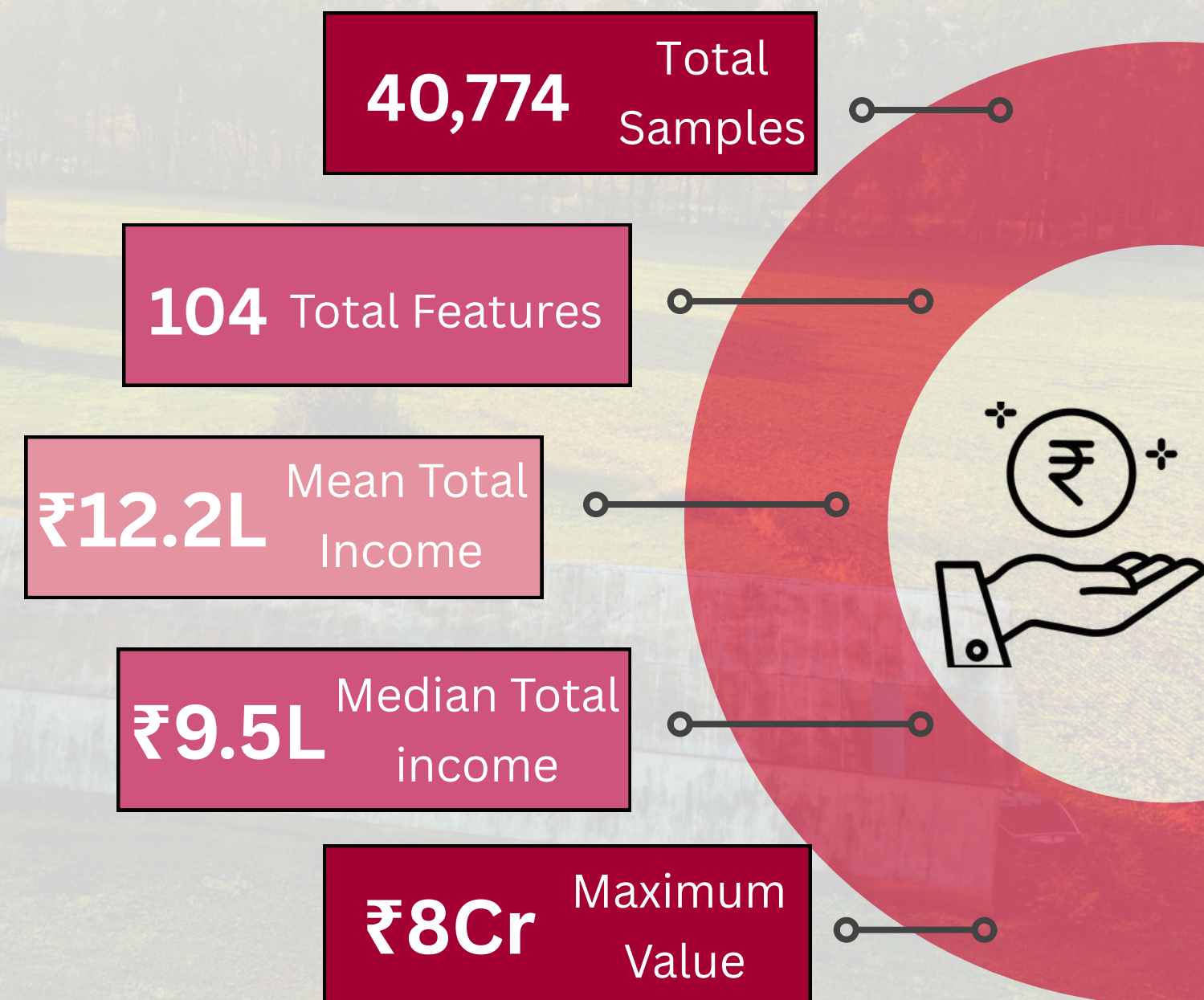


Exploratory Data Analysis : Nature of target variable

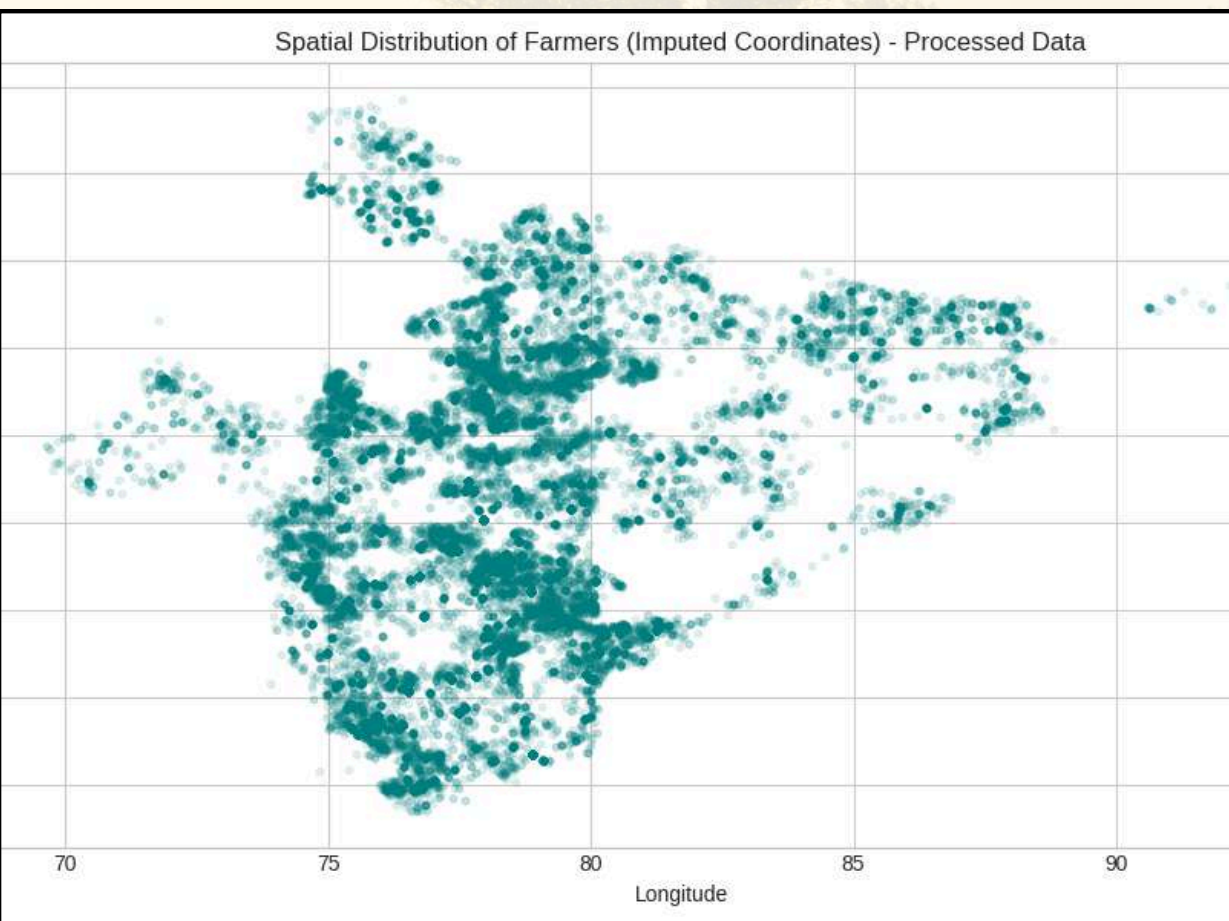


Target variable (Total Income) was heavily **right-skewed**

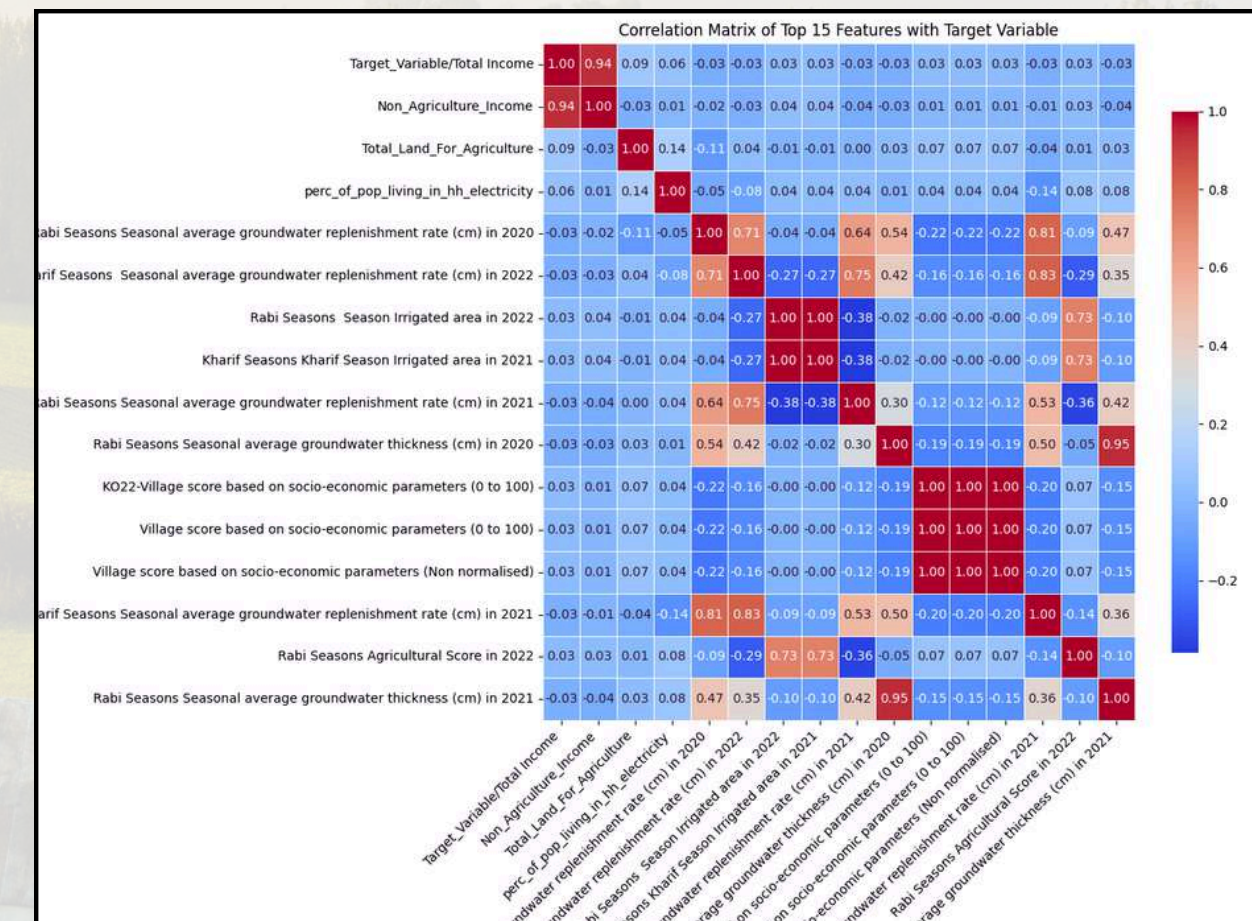
Log Transformation applied to normalize the distribution before training.



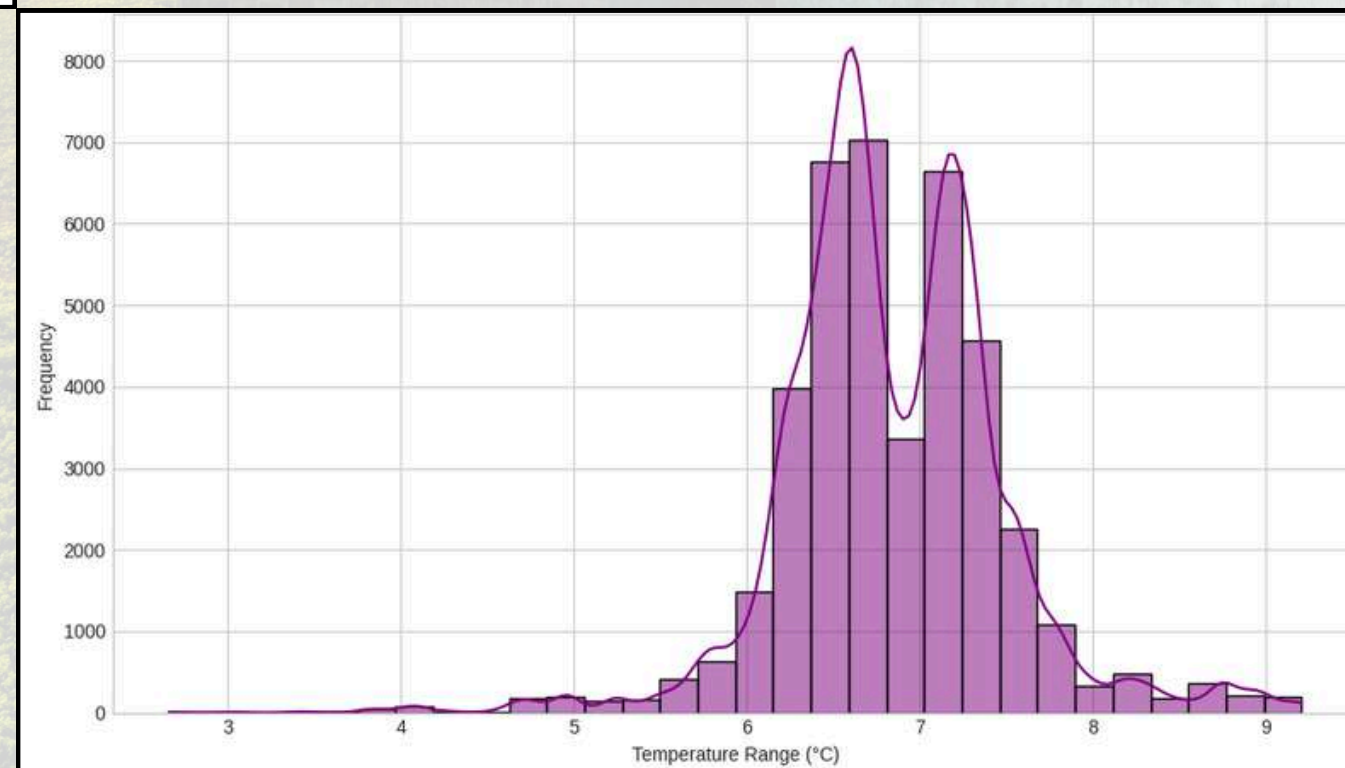
Exploratory Data Analysis : Geo- Economic Mapping



Temperature was mentioned as a min/max string. It was **parsed and separated** for the mathematical models to interpret and utilise it better



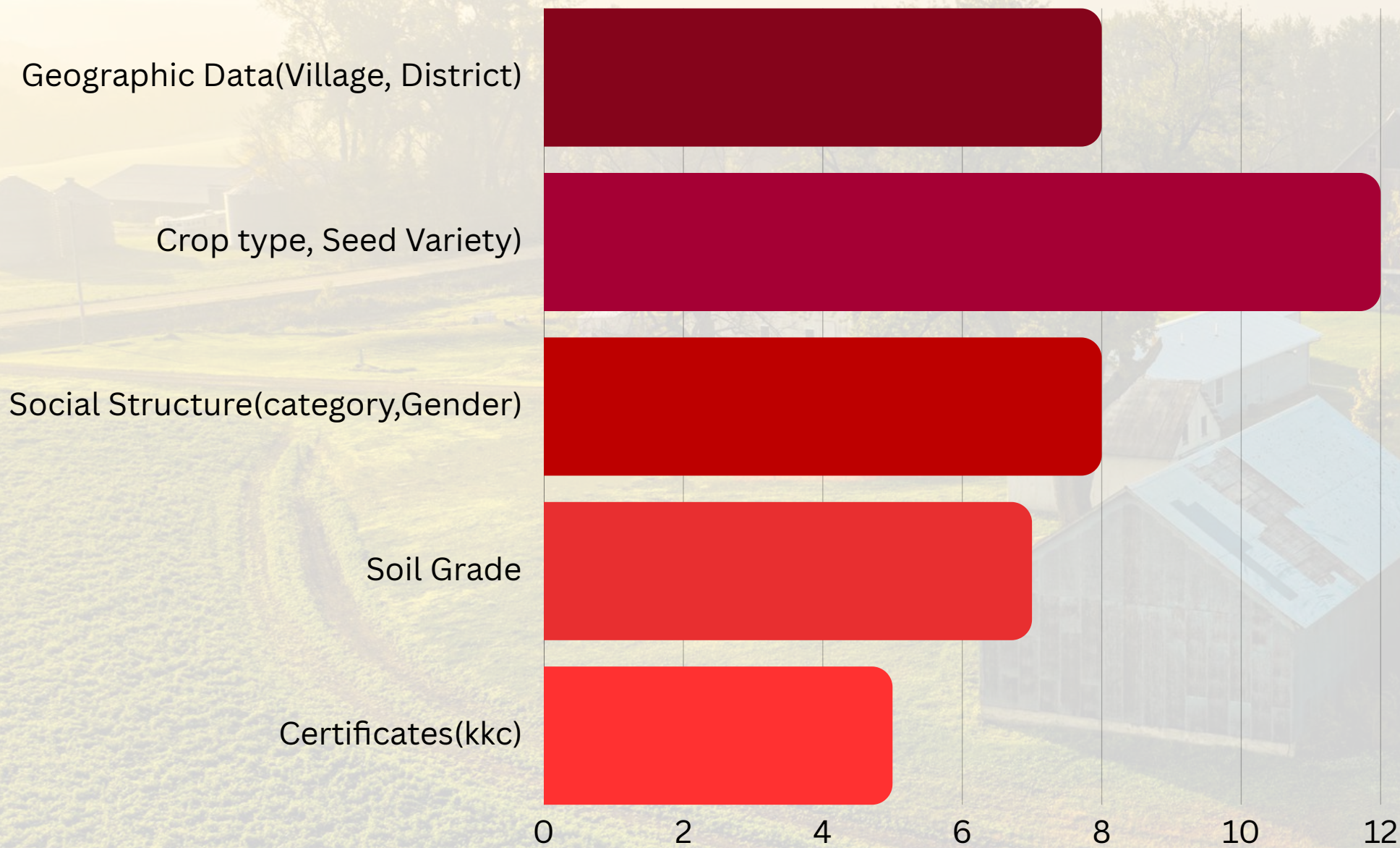
Coordinates parsed from Location string, **17,500+ missing values** imputed with district median covering a vast topology of India



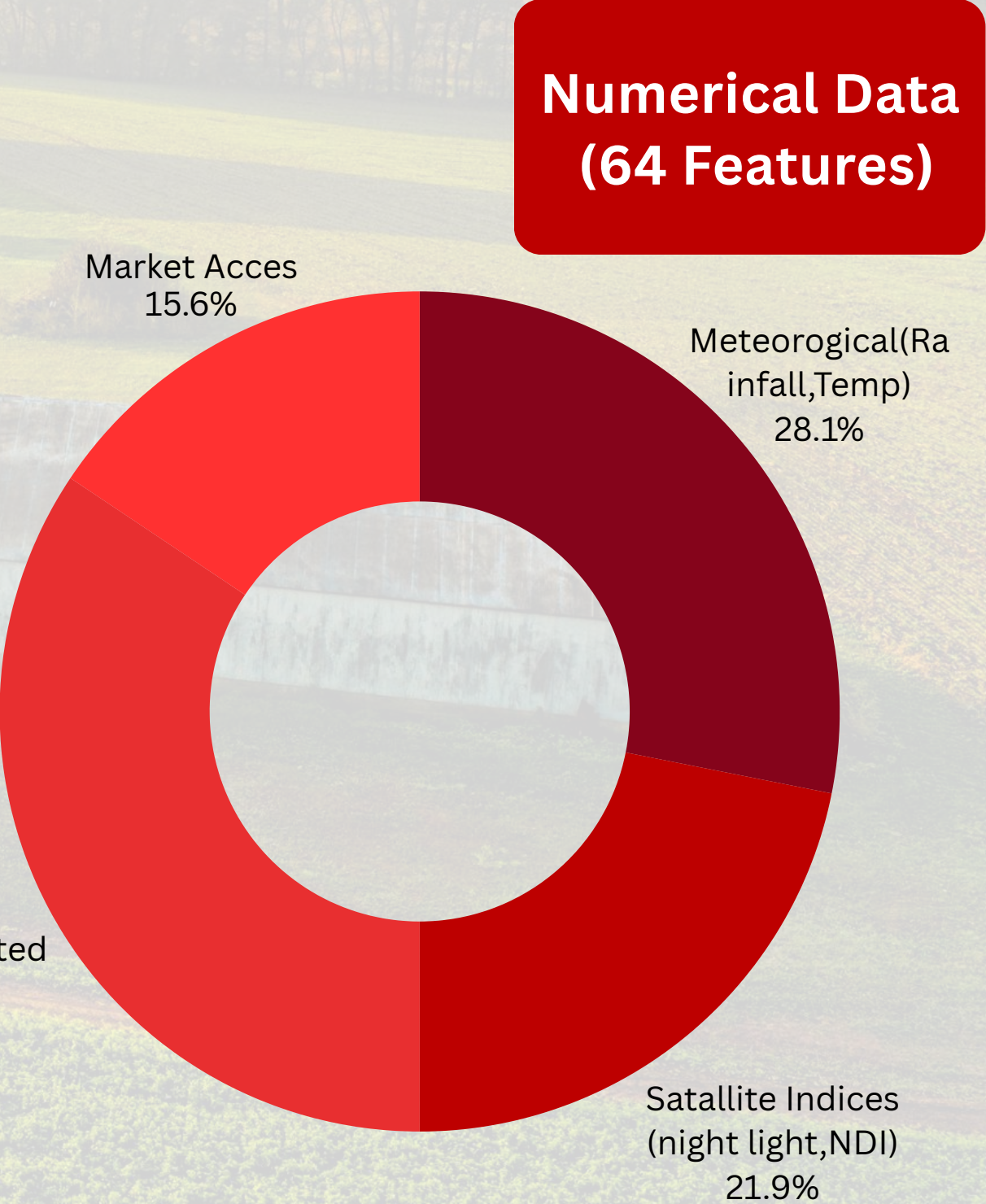
Correlation matrix of the top 15 features, most of which are **agricultural and seasonal** and a few are location-based



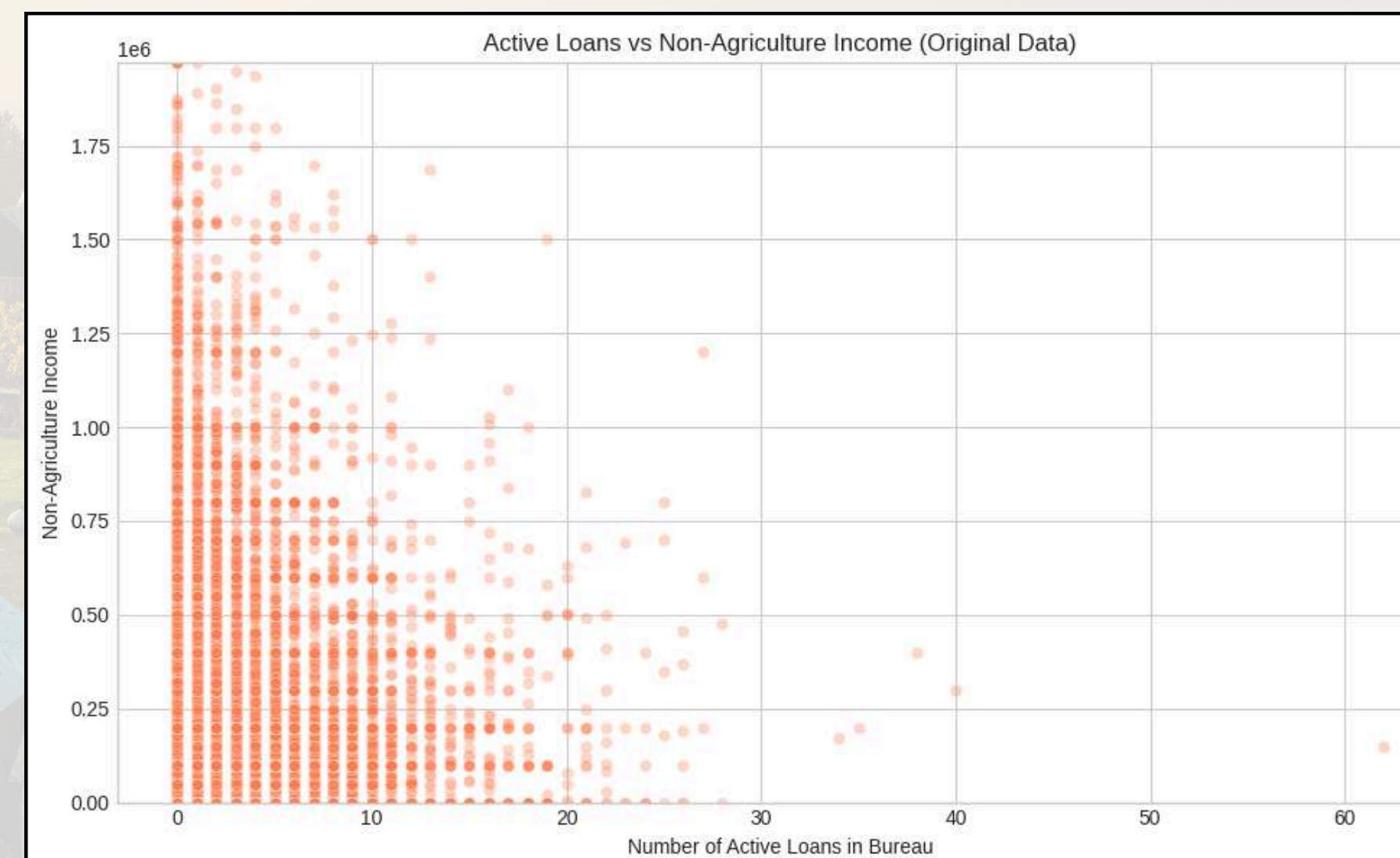
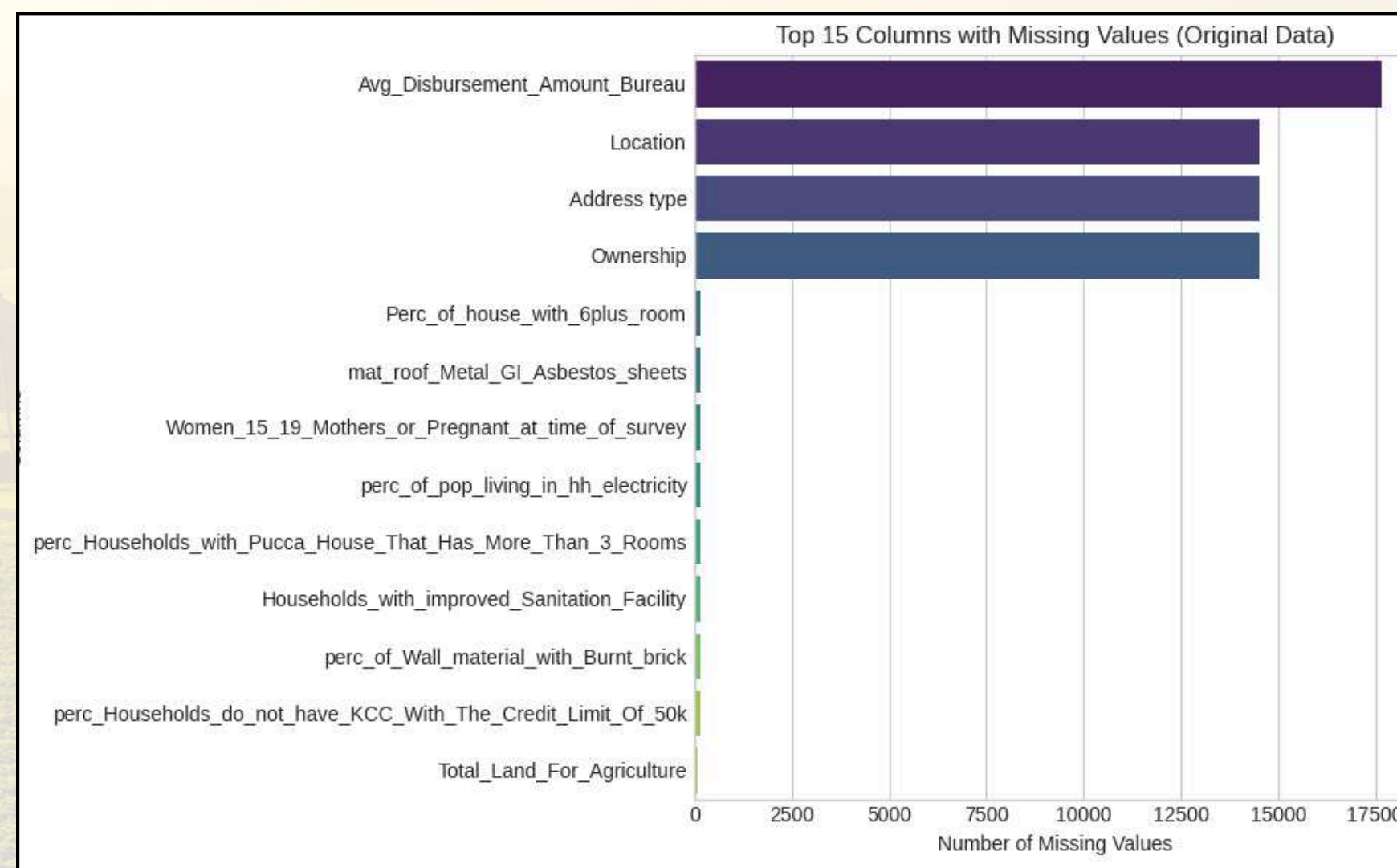
Exploratory Data Analysis : Type of Data Given



**Categorical Data
(12+28Features)**



Missing Data Evaluation : Nature of target variable



Address type, Ownership, Location, and Bureau Disbursement account for the bulk of missing data in absolute terms. These were **imputed** smartly so that there is no loss of data about the missing terms.

The number of active loans **exponentially decreases** with increase in **Non-agricultural income** which is understandable. The more the non-agricultural income, the **lesser** is the need for loans

Missing Data Evaluation : Differential Encoding



Differential Encoding transitions the model from analyzing static snapshots to evaluating temporal trajectories. Instead of treating 2020, 2021, and 2022 categorical data as independent labels, CDE isolates the "Delta" (the change in state) across years.

Strategic Rationale:



Momentum as a Signal:

Absolute status like "Average Village Score" is a baseline; the **change** in status like "Poor -> Average" represents economic **momentum**, which is a **higher-impact predictor** of income growth.



Infrastructure Proxy:

Transitioning categories in features like Type of Water Bodies or Soil Health Scores serves as a **proxy** for **infrastructure investment** or **land degradation** that static labels fail to surface.

$$\text{Differential Grade} = \text{Category Mean} - \text{Global Mean}$$



Noise Distillation:

Sequential agricultural data is often highly redundant. **Encoding** the "Difference" collapses three years of correlated data into a **single, high-variance Transition Index**, reducing **feature dimensionality** without losing historical context.



Error Reduction (MAPE):

By focusing on the **volatility of categorical states**, the models better identify the "Economic Resilience" of a farmer, directly addressing the **non-linear nuances** required to push MAPE below a threshold.



Model Formation : Models we tried out

We tried out several models, finetuned them and tried to get the best out of each one.
 The summary of the models we trained is:

XGBoost (All features)		XGBoost (with Top 35 features)	
n_estimators	1000	n_estimators	1000
learning rate	0.05	learning rate	0.05
max_depth	6	max_depth	6
sub_sample	0.8	sub_sample	0.8
column_sample_bytree	0.8	column_sample_bytree	0.8
objective	reg:absoluteerror	objective	reg:absoluteerror
random_state	42	random_state	42
n_jobs	-1	n_jobs	-1
Early stopping rounds	50	Early stopping rounds	50
MAPE : 0.1925 = 19.25%		MAPE : 0.1978 = 19.78%	

Random Forest Top 35 Features		CatBoost	
n_estimators	200	iterations	2500
max_depth	18	learning_rate	0.06
min_sample_leaf	4	depth	8
random_state	42	random_seed	42
n_jobs	-1	verbose	200
max_features	0.4		
MAPE: 0.1995 = 19.95%		MAPE: 0.2006 = 20.06%	

Light Gradient Boosting		Neural Network		
n_estimators	1000	Layer 1	128	relu
learnin_rate	0.05	Dropout 1	0.3	
max_depth	7	Layer 2	64	relu
num_leaves	60	Dropout 2	0.2	
subsample	0.8	Layer 3	32	relu
colsample_bytree	0.8	Dropout 3	0.1	
min_child_samples	20	Output Layer	1	linear
		epochs	150	
		batch_size	64	
MAPE: 0.19711 = 19.711%		MAPE: 0.2269 = 22.69%		

Testing and Selection : Why Tree-Based Boosting Model?



Sequential Error Correction - The Boosting Paradigm:



- Bagging (Random Forest) builds independent trees and averages outputs, without directly correcting errors
- Boosting builds trees sequentially, each one trained on residual errors of the previous model
- leads to focused, iterative error correction

Mastery of **Non-Linear** Agricultural Thresholds:



- Agricultural income does not scale in a straight, linear line. It is dictated by intersecting thresholds like Income drops IF Rainfall > 1000mm AND Soil is Sandy, UNLESS Irrigation is Present.
- Tree-based models natively map these complex, multi-variable interactions
- No requirement for us to manually engineer polynomial equations

Exceptional Handling of **High-Cardinality** Categoricals:



- One-Hot Encoding would expand 106 features into 10,000+ sparse columns, making the dataset inefficient and hard to train
- Linear and deep models depend on such encodings, leading to memory issues and poor handling of high-cardinality text
- CatBoost and LightGBM handle raw categories using techniques like Ordered Target Encoding

Built-in Resilience to Tabular Noise:



- Boosting models handle noisy, overlapping features using subsampling (e.g., ~80% data per tree) and L1/L2 regularization to reduce overfitting
- They filter out statistical noise during training instead of overreacting to it
- They gracefully handle missing values by learning default split directions, unlike neural networks which struggle with incomplete data



Boosting and Ensembling : Model Architecture

XGBoost - The Robust Optimiser

- Capitalizes on our Differential Grades.
- It uses strict mathematical regularization L1/L2 to rigorously map these numerical transitions without overfitting on noisy, high-income outliers.

LightGBM - The Pattern Hunter

- Leaf-Wise Growth.
- aggressively digs deep into the data
- finds highly specific, multi-variable interactions that standard models miss.

FINAL PREDICTION

CatBoost - The Categorical Anchor

- Natively excels with text/categorical variables.
- It utilizes Symmetric Trees which act as a conservative counterweight to noise.
- Ensuring stable predictions across diverse local geographies.

Why Ensembling?

Error Cancellation:

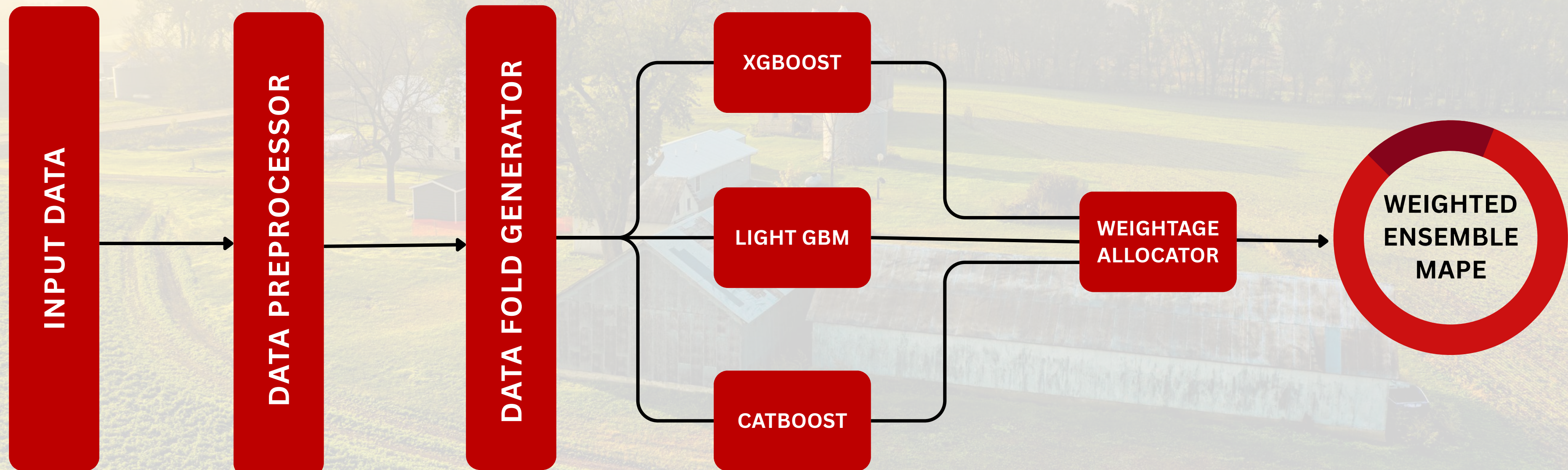
Every algorithm has inherent mathematical blind spots. Where LightGBM might over-predict a niche data pocket, CatBoost's conservative symmetry will under-predict it. Averaging them neutralizes the individual errors.

Stabilizing Volatility:

Agricultural income is highly susceptible to unpredictable real-world noise. An ensemble acts as a statistical shock-absorber, finalizing a prediction only when three distinct architectures reach a consensus.



The Final Architecture: Tri-Model Boosting Ensemble



Metric: MAPE: 18.544%



Model Evaluation : Output Analysis

Baseline Models
MAPE: 21-22%

K-folding Boosting

Independent Models
MAPE: 19.50-20%

Hypertuning Ensembling

Trimodel Ensemble
MAPE: 18.5%

Evaluation Metric
MAPE: 18.544%

	Count	MAPE	MAE	RMSLE
Bracket				
1. Moderate (<15 lakhs)	34507.0	16.81%	136659.15	0.2215
2. High (15-50 lakhs)	5782.0	27.40%	608403.12	0.4511
3. Very High (>50 Lakhs)	485.0	36.43%	4929030.31	0.8907



Custom Metric : Root Mean Squared Log Error

$$\text{RMSLE} = \sqrt{\frac{1}{n} \sum (\log(y + 1) - \log(\hat{y} + 1))^2} = 0.2825$$

We chose RMSLE because standard metrics like MAE and MAPE fail to capture the reality of agricultural income. RMSLE solves three problems:

Neutralizes Outliers	Mathematical Stability	Business Reality
- Income is massively right-skewed. MAE would let a ₹5 Lakh error on a wealthy corporate farmer completely overshadow a ₹20,000 error on a smallholder - RMSLE measures the ratio of the error, forcing the model to care equally about all income brackets	- If a farmer reports near-zero income due to crop failure, standard MAPE mathematically explodes to infinity and crashes the model - RMSLE naturally handles zeros, ensuring stable training and exception handling	- MAPE inherently trains models to under-predict to minimize percentage errors. RMSLE does the opposite penalizing under-estimating slightly more than over-estimating - Under-estimating a farmer's wealth leads to unfairly denied loans. RMSLE ensures our model doesn't discriminate against them



Thank You

Special Thanks to our mentors - Arnav Gawade, Soham Agarwal and Chirag Khandelwal for their invaluable guidance throughout the hackathon

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